

Answer Key

Task 1: Holding Period Return

$HPR = (\text{Ending value} - \text{Beginning value} + \text{Dividends}) / \text{Beginning value}$

$HPR = (32,400 - 30,000 + 600) / 30,000 = 3,000 / 30,000 = 10\%$

- HPR represents the total return earned over the period, combining both the capital gain and the income received.
- HPR alone doesn't tell us whether the return was good, whether it compensated for the risk taken, or whether it beat inflation or a benchmark — it's a starting point, not a full performance evaluation.

Task 2: Arithmetic vs. Geometric Returns

Arithmetic average = $(12\% - 5\% + 8\%) / 3 = 5\%$

Geometric average = $[(1.12)(0.95)(1.08)]^{(1/3)} - 1 \approx 4.74\%$

- The geometric average is lower because it accounts for compounding: the -5% loss applies to a base that already includes the Year 1 gain, so it erodes more value than the arithmetic average implies.
- Compounding matters for long-term investors because actual wealth grows multiplicatively, not additively; the geometric average is the rate that actually reflects how their money grew.

Task 3: Risk-Adjusted Performance

Sharpe A = $(11\% - 3\%) / 15\% \approx 0.53$

Sharpe B = $(9\% - 3\%) / 10\% = 0.60$

Treynor A = $(11\% - 3\%) / 1.2 \approx 6.67$

Treynor B = $(9\% - 3\%) / 0.7 \approx 8.57$

- **Portfolio B performs better on both measures**, despite its lower raw return. It earns more excess return per unit of risk, whether risk is measured as total volatility (Sharpe) or market risk alone (Treynor).
- Sharpe and Treynor can produce different rankings when a portfolio's total risk and systematic risk don't move together. For example, a portfolio with high unsystematic (diversifiable) risk but low beta could rank well on Treynor but poorly on Sharpe. In this case both rankings agree, but that won't always happen.

Task 4: Fisher Equation and Real Returns

Real return = $(1 + \text{Nominal}) / (1 + \text{Inflation}) - 1 = (1.08 / 1.05) - 1 \approx 2.86\%$

- Real returns matter more for long-term wealth preservation because what ultimately matters is purchasing power, not the dollar amount. A portfolio can post a healthy

nominal return and still lose ground economically if inflation erodes it — nominal returns alone can overstate how much wealth is actually being built.

Task 5: Downside Risk

- A much lower Modified Sharpe Ratio and higher CVaR, despite a similar plain Sharpe Ratio, suggests the fund has significant **tail risk**: its losses in bad scenarios are more extreme than a normal-distribution-based measure like standard deviation would suggest. The ordinary Sharpe Ratio is being flattered by ignoring this.
- Downside risk matters because investors care asymmetrically about losses, not volatility in general; two funds can look equally "risky" by standard deviation while one is far more exposed to a severe crash.
- Investors may prefer downside-focused measures (like CVaR or Modified Sharpe) in volatile markets because ordinary risk measures assume a symmetric, well-behaved return distribution, an assumption that tends to break down precisely when markets are stressed and tail events are most likely.